PEARL: An Open Source Python Tool for Lifetime Estimation of IGBT–Diode Power Modules

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***Abstract*—This paper presents semiconductor focused implementation of PEARL (Power Electronics Age and Reliability), an open source, Python based framework for lifetime assessment of power electronic components in inverter based resources. It couples electrical loss modelling, dynamic junction temperature estimation and power cycling lifetime estimation into a unified and reproducible workflow, enabling mission profile based reliability analysis of IGBT modules with its anti-parallel diodes under realistic operating conditions. In this framework, device level electrical losses are first computed and mapped to junction temperature trajectories using multi stage Cauer type RC thermal networks, after which the resulting temperature cycles are evaluated using a power cycling lifetime model combined with Miner's cumulative damage rule to estimate semiconductor's lifetime. Case studies demonstrate the capability of the framework to (i) quantify the impact of reactive power operation on IGBT–diode power modules lifetime and (ii) identify heat sink configurations that keep junction temperatures of IGBT-diode power modules within safe limits under peak conditions.**

***Index Terms*—Power electronics reliability, IGBT-diode module, electro-thermal modelling, mission profile, lifetime estimation**

I. Introduction

The ongoing transition toward converter dominated power systems has made the reliability of power electronic converters central to grid resilience as they increasingly form the interface between generation, storage and the grid. In solar PV plants, inverters account for over 60% of system failures, corresponding to an estimated $2.5 billion in annual losses [1], and in wind turbines, power electronic converters remain a primary source of downtime [2]. At the same time, inverter based resources (IBRs) are no longer passive grid followers but are increasingly required to provide grid critical services such as voltage regulation, reactive power support and fault ride through [3]. These services impose additional electrical and thermal stresses on semiconductor devices within converters, accelerating aging and amplifying reliability concerns [4], [5], [6]. Collectively, these observations highlight power electronics as a key reliability bottleneck in the energy transition, underpinning essential grid support functions and challenging long term grid planning and asset economics [1], [7]. As the deployment and market participation of IBRs continue to expand [8], robust reliability assessment of power electronic components becomes essential for long term asset performance and system level planning.

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At the heart of these reliability challenges lie the IGBT-diode power modules that form the switching core of modern converters. The dominant failure mechanisms in IGBT-diode power modules arise from deterministic wear out processes driven primarily by thermo-mechanical fatigue [9]. Field data indicate that temperature related loading accounts for more than 55% of failures in power-electronic systems [10]. Such failures are induced by repetitive junction temperature cycling caused by fluctuating electrical loading and environmental conditions, which progressively degrade bond wires, solder layers and interconnect structures over time [9].

A. Identified Research Gap

The growing penetration of power electronic converters exposes a key tooling gap that there is still no unified, transparent and reproducible framework that maps realistic, time varying mission profiles to quantitative component lifetime consumption. As summarized in Table I, existing reliability approaches are fragmented such as high-fidelity methods (e.g., FEM and accelerated aging tests) provides detailed insight but are too computationally expensive for long duration, system level mission profiles [11], while manufacturer tools are convenient and device calibrated but closed source, limited in scope and difficult to integrate into broader operational studies. Consequently, despite the central role of power electronics in providing grid critical functions, an accessible mission profile driven reliability framework remains missing.

B. Paper Contribution

This paper introduces PEARL, an open source, physics based electro-thermal reliability framework for power-electronic systems, with a first implementation focused on IGBT-diode power modules. The framework links time varying mission profiles, operating conditions, design configurations and component specifications to semiconductor lifetime consumption by unifying electrical loss modelling, dynamic junction temperature estimation and power cycling lifetime estimation.

C. Paper Organization

Section II presents the proposed methodology, Section III describes the case studies used for evaluation and Section IV discusses the corresponding results and key findings. Finally, Section V concludes the paper, while Section VI outlines directions for future work.

TABLE I: Comparison of available commercial and open-source tools for power electronics lifetime / RUL / electro thermal analysis.

Tool	Type	Main capabilities	Strengths	Weaknesses / watchouts
CeRULEo	Open-source / Python	Train/evaluate RUL ML models	Good for ML benchmarking	No device to mission physics chain
ProgPy / NASA libs	Open-source / Python	PHM / prognostics components	Useful for RUL algorithm validation	Not power electronics specific
Infineon IPOSIM	Manufacturer-proprietary	Device specific lifetime estimation	Easy, vendor calibrated	Closed-source; limited to Infineon parts
Semikron / SemiSel	Manufacturer-proprietary	Loss/thermal calculators; lifetime curves	Manufacturer recommended curves	Lacks system level RUL integration
PLECS (Plexim)	Commercial (simulation)	Electrical-thermal co-simulation	Well integrated; widely used	Requires custom RUL integration
ANSYS (Icepak / Sherlock)	Commercial (FEM)	3D CFD/thermal + physics of failure	High-fidelity FEM, thermomechanics	Heavy, costly; not real-time
COMSOL	Commercial (FEM)	3D multiphysics thermo-mechanical	Excellent for hotspot/stress validation	Slow; not for real time RUL
MathWorks RUL Toolbox	Commercial (toolbox)	Data driven, Simulink integration	Rich ML + model hybrid tools	License cost; needs physics integration

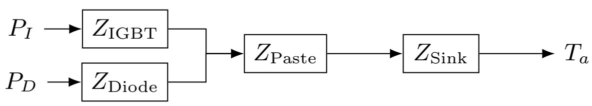

Fig. 1: Cauer type thermal network of IGBT–diode module.

II. Methodology

As described previously, the IGBT-diode power module implementation of PEARL combines electrical loss, thermal and degradation models into a single reproducible lifetime assessment pipeline, which is described step by step in the remainder of this section.

IGBT and diode power losses are computed from the mission profile operating points and the inverter configuration by first reconstructing the instantaneous device current waveforms through the IGBT and its anti parallel diode. Based on these currents, conduction and switching losses are evaluated using the established formulations in [4], yielding time series power loss profiles for the IGBT and diode across the mission profile. All required loss model parameters are extracted directly from the manufacturer datasheets of the selected semiconductor module.

For the thermal analysis, the time varying power loss profiles of the IGBT and diode are applied to a multi stage Cauer type RC thermal network (Fig. 1) that represents the heat flow path from the device junctions to the ambient. In this network, the junction to case thermal dynamics of the IGBT and diode are modeled by the impedances Z_{IGBT} and Z_{Diode}, respectively, while the shared case to ambient path is represented by Z_{Paste} and Z_{Sink}, capturing heat transfer through the thermal interface material and the heat sink to the ambient T_a. The RC parameters are derived from manufacturer provided transient thermal impedance data, allowing the model to capture relevant thermal time constants and the thermal coupling between both devices through the common downstream path. Solving the resulting coupled system of ordinary differential equations yields the IGBT and diode junction-temperature trajectories under time varying losses and ambient conditions, which form the input for the subsequent lifetime assessment.

The resulting junction temperature trajectories are post processed using rainflow counting to extract thermal cycles (amplitude, mean temperature and count). These cycles are then mapped to cycle to failure using the power cycling lifetime model in [12] and cumulative lifetime consumption is computed via Miner's linear damage rule for the considered mission profile. This enables the lifetime estimation of IGBT and diode under the considered mission profiles.

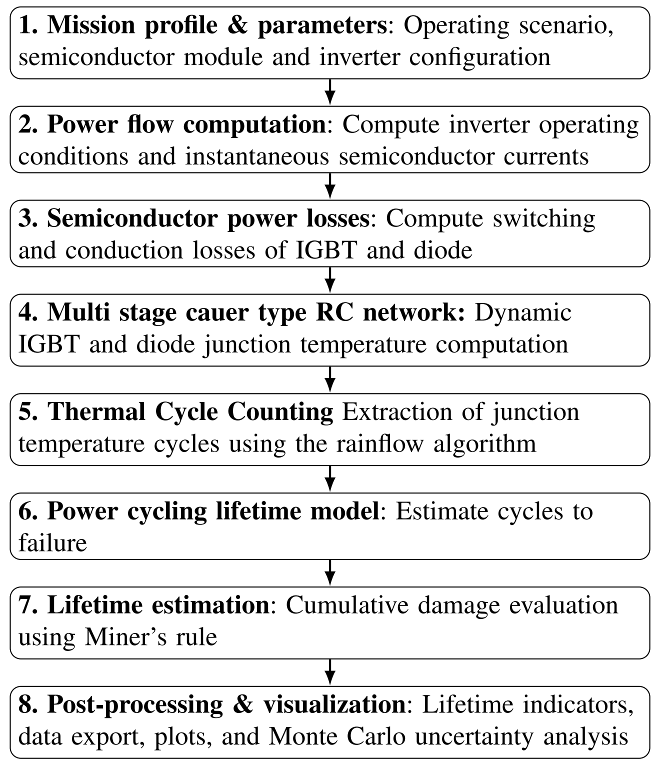

Fig. 2: Methodological workflow.

Fig. 2 provides a visual overview of the framework pipeline. Further model explanations are provided in *Model_documentation_PEARL_IGBT_and_Diode.pdf* in the *IGBT and Diode* folder of the PEARL GitHub repository [13], covering the electrical loss, thermal and lifetime assessment models. In practice, the semiconductor implementation of PEARL requires only a small set of open and readily available inputs such as manufacturer datasheet parameters for the selected IGBT, diode, thermal interface material and heat sink, together with the inverter configuration and time-varying active and reactive power profiles.

III. Case Study

A. Case Study 1: Reactive power impact on power module

Motivated by the growing requirement for power converters to provide reactive power support, this case study quantifies how reactive power provision by inverter based resources affects the lifetime of IGBT-diode power modules. The simulation setup is illustrated in Fig. 3. In this scenario, the IBR operates at rated apparent power for a duration of 1 hour, with constant power factor values between 0 and 1 under both inductive and capacitive operating conditions. The mission profile is cyclically repeated over the service lifetime to evaluate the influence of reactive power on semiconductor lifetime. Although a simplified operating condition with constant rated power and constant ambient temperature is considered for demonstration purposes, the framework is fully capable of handling realistic, long duration mission profiles with time varying active power, reactive power and ambient temperature.

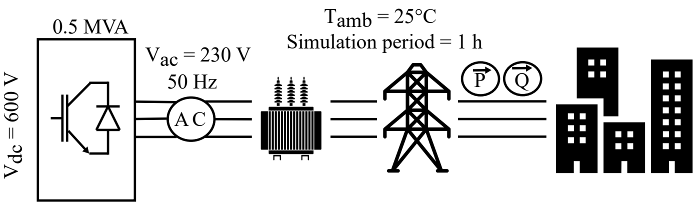

Fig. 3: Case study framework

The inverter is modeled as a three-phase converter rated at 0.5 MVA, with an AC side phase RMS voltage of 230 V and a DC link voltage of 600 V. It operates at a modulation index of 1 using space vector modulation (SVM) with a switching frequency of 10 kHz and a grid frequency of 50 Hz. Each switch position is modeled as a power semiconductor module comprising ten identical IGBT with anti-parallel diode pairs connected in parallel to achieve the required current rating.

The semiconductor device considered is the 600 V, 50 A IKW50N60H3 IGBT with integrated anti parallel diode from Infineon [14]. The junction to case transient thermal impedance characteristics of both the IGBT and diode are extracted and implemented as multi stage Cauer type RC networks to capture dynamic thermal behavior. In addition to thermal parameters, switching and conduction characteristics are obtained from the datasheet to parameterize the electrical loss model. The case to sink interface is modeled using Thermal Grizzly Kryonaut thermal paste with a thermal resistance of 0.0032 K/W [15]. The heat sink is represented by the aluminum Cooling Innovations 3-121211U device, with a specific heat of 900 J/kg K and thermal resistance of 2 K/W at air speed of 1 m/s [16]. All numerical parameter values used in this case study are listed in *Input_parameters.py* in the *IGBT and Diode* folder of the PEARL GitHub repository [13].

B. Case Study 2: Heat sink selection for power module

During inverter design, semiconductor devices must be paired with an adequately sized heat sink to ensure that the junction temperature does not exceed the manufacturer specified limit. In this study, the IKW50N60H3 IGBT-diode module [14] has a maximum allowable junction temperature of 175°C, which defines the thermal design constraint. To satisfy this requirement, heat sinks from the Cooling Innovations aluminum series 3-1212 [16] are evaluated. These heat sinks differ in height, weight and thermal resistance as a function of air speed, as summarized in Table II. For the thermal design assessment, a constant ambient temperature of 25°C is assumed and the inverter is operated at rated apparent power with pf $= 1$, and this operating point is used to determine the heat sink configuration required to ensure safe semiconductor operation.

TABLE II: Heat sinks specifications

P/N	Area[mm^2]	Height[mm]	Weight[g]	Thermal Resistance [°C/W]			
				1m/s	2m/s	3m/s	4m/s
02U	967.21	5.1	7.0	11.6	7.34	5.39	4.27
04U	967.21	10.2	9.7	5.46	3.20	2.28	1.79
06U	967.21	15.2	12	3.62	2.08	1.48	1.16
08U	967.21	20.3	15	2.72	1.55	1.11	0.88
10U	967.21	25.4	18	2.20	1.25	0.90	0.72

IV. Results and discussion

Figures 4 illustrate the internal electro-thermal computation chain of the framework for a certain operating point. For clarity, only one fundamental grid cycles (20 ms at 50 Hz) is shown.

As expected, the model reconstructs the instantaneous semiconductor currents from the converter operating conditions in a physically consistent manner, as shown in Fig. 4(a). The current is shared between the IGBT and the anti-parallel diode according to the phase relation between voltage and current: when they are in phase ($pf = 1$), the IGBT conducts most of the current, whereas with increasing phase shift($pf = 0$), the diode conduction becomes more pronounced. This confirms that the model captures the expected redistribution of instantaneous current between the semiconductor devices.

Based on the IGBT-diode currents along with other parameters, the instantaneous power losses of the IGBT and diode are calculated, as shown in Fig. 4(b). The total semiconductor loss consists of conduction and switching components. Conduction losses arise from the current flowing through the device during its conduction interval and are determined by the device voltage and current magnitude, while switching losses occur during turn-on and turn-off transitions due to the overlap of voltage and current.

These time-varying power losses are applied to the multi-stage Cauer-type thermal network, which represents the transient junction temperatures of the IGBT and diode as shown in Fig. 4(c). At $pf = 1$, most of the current flows through the IGBT, resulting in higher IGBT losses and a noticeable temperature rise, while the diode temperature remains nearly constant due to its negligible losses. At $pf = 0$, the IGBT still dissipates more power overall, while the diode dissipates less but exhibits a higher temperature. This is due to the higher thermal resistance of the diode (1.05 K/W) compared to the IGBT (0.45 K/W) [14], meaning that for similar power dissipation the diode experiences a larger temperature rise. Additionally, the temperature peaks occur slightly after the power-loss peaks due to thermal inertia, and the temperature decreases gradually even after conduction stops because the device requires time to cool through the thermal path.

Fig. 5 shows the junction-temperature evolution of the IGBT and diode over the 1-hour simulation. As heat accumulates in the thermal network, the temperatures converge to a steady-state value, which remains constant under the fixed operating conditions of Case Study 1 and is used for the subsequent lifetime calculation.

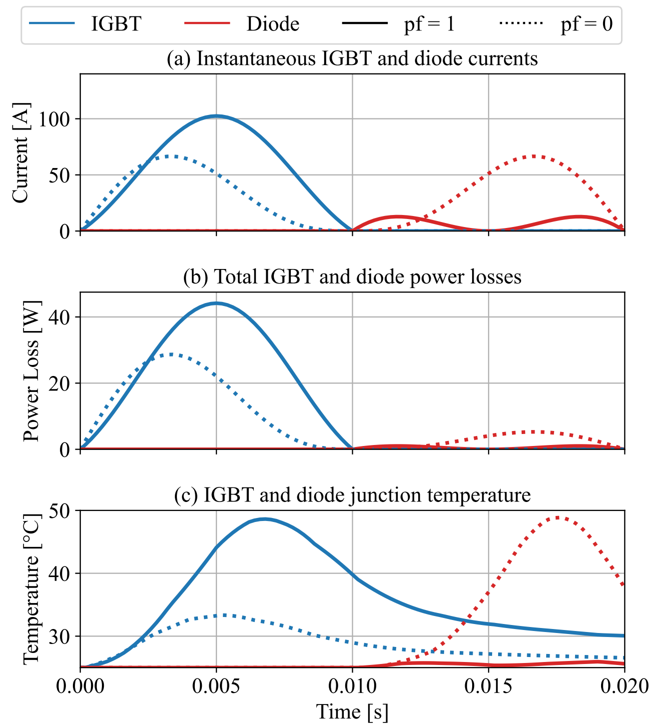

Fig. 4: Electro–thermal response of the IGBT-diode module: (a) Instantaneous device currents, (b) Total power losses, (c) Junction temperature.

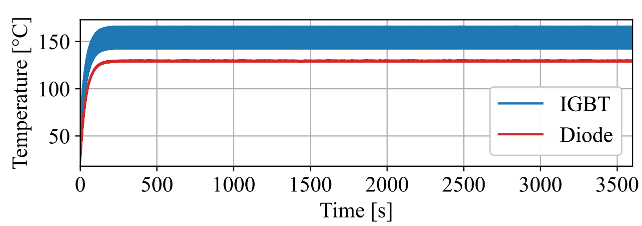

Fig. 5: Long-duration junction temperature of IGBT and diode

A. Case Study 1: Reactive power impact on power module results

Figure 6 summarizes the influence of power factor on estimated lifetime of the IGBT-diode power module. The inverter is simulated at constant apparent power while the power factor is varied from 0 to 1 for both inductive and capacitive conditions, resulting in 20 operating scenarios. For each case, the average device current, total losses, junction temperature and estimated lifetime are extracted for comparison.

As shown in Fig. 6(a), the average device currents redistribute between the IGBT and its anti-parallel diode as the power factor varies, while the total phase current remains constant due to constant apparent power operation. At unity power factor, voltage and current are aligned and most of the current flows through the IGBT with limited diode conduction. As the phase angle increases toward 90° ($pf \rightarrow 0$), diode conduction intervals increase, leading to a complementary redistribution of current between the two devices.

This redistribution of device current directly influences the semiconductor losses, as shown in Fig. 6(b). Near unity power factor, the IGBT conducts for longer intervals, leading to higher average IGBT losses and reduced diode dissipation, whereas as the power factor approaches zero, a larger fraction of the current returns through the diode, increasing its conduction losses and reducing those of the IGBT. The loss trends are not perfectly symmetric because the two devices exhibit different electrical characteristics. The IGBT incurs both conduction and switching losses, including turn-on and turn-off energy components that scale nonlinearly with current, while the diode losses are primarily governed by conduction and reverse-recovery effects. Consequently, the total power dissipation, defined as the sum of switching and conduction losses, evolves differently for the two devices as the power factor varies. Further details on the underlying loss modeling and device physics are provided in *Model_documentation_PEARL_IGBT_and_Diode.pdf* [13].

This redistribution of current directly affects the semiconductor losses, as shown in Fig. 6(b). Near unity power factor, higher IGBT current results in increased IGBT losses and reduced diode losses, while as the power factor approaches zero, the diode current increases, raising diode losses and reducing those of the IGBT. However, the loss trends are not solely determined by current magnitude, since the IGBT experiences both conduction and switching losses, whereas the diode losses are mainly governed by conduction and reverse-recovery effects.

The resulting thermal response due to the device power losses is shown in Fig. 6(c). As is well known, the junction-temperature rise is approximately proportional to the dissipated power, with the proportionality factor given by the thermal resistance. For the considered module, the thermal resistance is $0.45\ \mathrm{K/W}$ for the IGBT and $1.05\ \mathrm{K/W}$ for its antiparallel diode [14]. Hence, each watt of dissipated power results in about $0.45\ \mathrm{K}$ and $1.05\ \mathrm{K}$ temperature rise for the IGBT and diode, respectively. However, the IGBT and diode are thermally coupled through the common case, thermal paste, and heat sink (see Fig. 1). Therefore, the junction temperature of each device is influenced not only by its own losses but also by the losses of the other device. At $\mathrm{pf} = 1$, the IGBT exhibits higher power losses, leading to a higher junction temperature. Although the diode losses are lower in this condition, its larger thermal resistance results in a comparatively stronger temperature rise per watt, and it is additionally heated by the elevated case temperature caused by the IGBT losses. As the power factor approaches zero, the IGBT losses decrease while the diode losses increase. Nevertheless, the strong reduction in IGBT heating lowers the common case temperature, which partially offsets the diode's increasing self-heating. At $\mathrm{pf} \approx 0$, where both devices dissipate comparable power, the diode reaches a higher junction temperature due to its significantly larger thermal resistance.

Finally, the estimated lifetime of IGBT and diode is illustrated in Fig. 6(d). THe lifetime of IGBT and diode is directly proportional to its junction temperature but both IGBT and diode will be affected differents as their parameter are different. One can clearly see a reverse trend at power factor close to unity igbt lifeitme is less and for diode is more and as we go close to power factor 0 igbt life is more and diode is less. The trend is not that simple to see as one has to also consider igbt and diode coupling and heat transfor but still. These results demonstrate that, despite constant apparent power operation, reactive power provision substantially alters device-level thermal stress and lifetime consumption.

~~Finally~~, the estimated lifetimes of the IGBT and diode are shown in Fig. 6(d). Since device lifetime is strongly influenced by junction temperature, the different thermal behaviors of the IGBT and diode lead to opposite lifetime trends. It should be noted that the impact of temperature on lifetime differs for the IGBT and diode due to their different lifetime model parameters, and the trends are slightly affected by the thermal coupling between the two devices. Near unity power factor, higher IGBT temperatures lead to reduced IGBT lifetime while the diode lifetime is increased. As the power factor approaches zero, the increasing diode temperature reduces its lifetime while the IGBT lifetime improves. These results show that, even under constant apparent power operation, reactive power provision significantly alters the thermal stress and lifetime consumption of the semiconductor devices.

Figure 6(d) presents the individual lifetimes of the IGBT and diode. However, since failure of either device leads to failure of the entire semiconductor module, the effective module lifetime is determined by the minimum of the two. Accordingly, Fig. 7 shows the module lifetime as a function of power factor, obtained by taking the lower value between the IGBT and diode lifetimes.

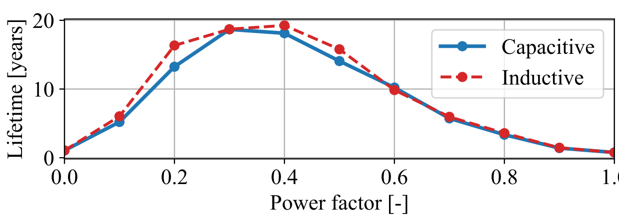

Fig. 7: Estimated lifetime of the semiconductor power module

It can be observed from Fig. 6 that inductive and capacitive operation at the same power factor produce nearly identical results. The small deviations visible in Fig. 6(c) and (d) arise from the numerical solution of the Cauer thermal model using ordinary differential equations and solver tolerances. Although higher numerical accuracy could reduce these deviations, it would significantly increase the computational effort therefore, a balanced setting between accuracy and computation time was selected.

B. Case Study 2: Heat sink selection for power module results

Fig. 8 shows the maximum junction temperature (the higher of the IGBT and diode temperatures) for different heat sink configurations. The heat sinks are evaluated for air velocities between 1 m/s and 4 m/s, which affect the heat sink thermal resistance. Smaller heat sinks (02U, 04U) have higher thermal resistance due to their limited surface area (see Table II),

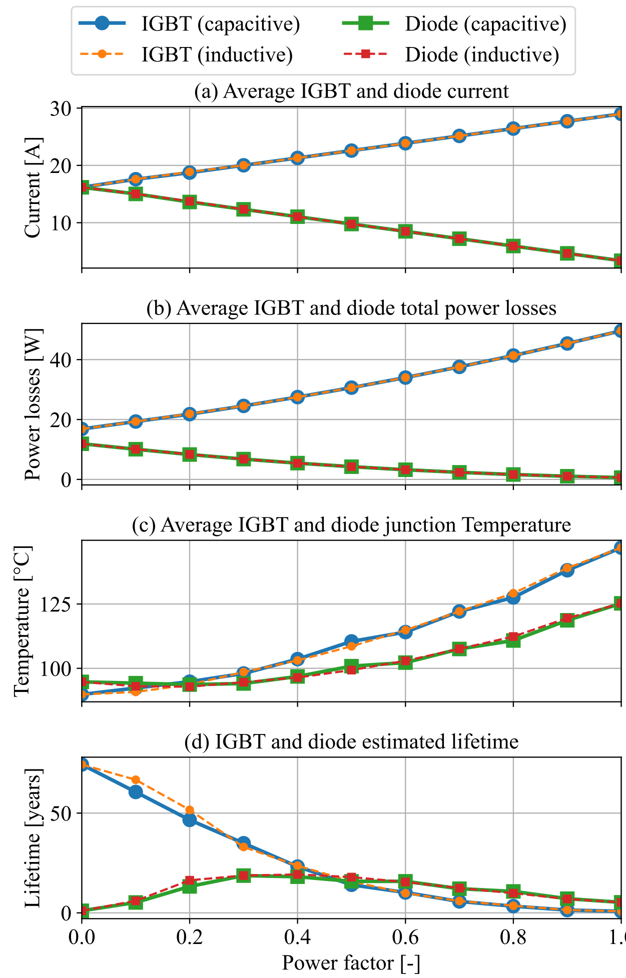

Fig. 6: Influence of power factor on the electro-thermal behavior and lifetime of the IGBT-diode power module: (a) average device current, (b) average total semiconductor power losses, (c) average junction temperature and (d) estimated lifetime.

resulting in higher junction temperatures that can exceed the allowable limit of 175°C. Increasing the heat sink size (06U, 08U, 10U) reduces thermal resistance, improves heat dissipation and keeps the junction temperature within the safe operating range.

Figure 8 illustrates the trade off between heat sink size and the required air flow for thermally safe operation. Smaller heat sinks require higher air velocities to keep the junction temperature below the allowable limit. For example, the 04U heat sink operates safely only at around 4 m/s, but its compact height (10.2 mm) results in a small volume of about $9865.54\ \mathrm{mm}^3$, roughly 2.5 times smaller than the larger 10U heat sink ($24567.13\ \mathrm{mm}^3$), which can maintain safe temperatures at air speeds slightly above 1 m/s. This highlights the design trade off that is compact heat sinks save space but require stronger forced cooling implying larger fans and increased auxiliary power consumption, while larger heat sinks reduce airflow requirements and auxiliary power consumption.

With this framework, such design decisions can be supported by quantitatively evaluating the thermal feasibility of different configurations. Although practical inverter design involves many additional considerations, semiconductor implementation of PEARL provides a physics-based assessment of semiconductor thermal behavior and helps identify which combinations of heat-sink size, cooling conditions and operating points are thermally viable.

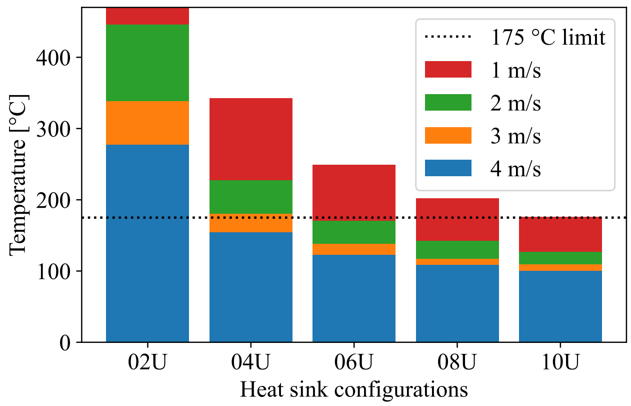

Fig. 8: Junction temperature for different heat sink types.

V. Conclusion

This paper presents the semiconductor focused implementation of PEARL, which integrates electrical loss modeling, dynamic junction-temperature modeling and power cycling lifetime estimation within a unified workflow for IGBT-diode power modules. Using parameters derived directly from manufacturer datasheets, the framework translates mission profile operating conditions into cumulative lifetime consumption, enabling systematic evaluation of how operating conditions affect semiconductor reliability.

Two case studies were conducted to demonstrate the capabilities of the framework. The first case study examined the impact of reactive power operation on semiconductor lifetime and showed that, even at constant apparent power, variations in power factor redistribute can lead to different lifetime for IGBT and diode. The second case study evaluated the influence of heat sink size and air flow conditions on the maximum junction temperature of the semiconductor module, illustrating how such analysis can support informed thermal design choices by balancing heat sink size, available space and cooling requirements during the early stages of inverter development.

Overall, the results demonstrate that the PEARL semiconductor focused implementation provides a transparent framework for linking converter operating conditions, design parameters, component specifications and semiconductor lifetime estimation, enabling early stage evaluation of reliability implications for different design choices.

VI. Future Outlook

The present work focuses on the semiconductor level implementation of PEARL. Future developments will extend the framework to additional power electronic components such as DC link capacitors, gate driver circuits, LCL filters etc. By incorporating lifetime models for these components, PEARL can be expanded toward system level reliability analysis of grid connected converters. In the long term, the framework aims to evolve into a flexible reliability assessment platform capable of modeling diverse power-electronic-based assets such as STATCOMs, HVDC converters, and other converter dominated infrastructure, enabling evaluation of grid wide reliability impacts under varying operating conditions.

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